GlycoTwin-AI: End-to-End Multimodal Digital Twin Architecture

Happiest Health Reimagining & Reforming Healthcare Summit 2026 • Proactive Diabetes Management in India

1. DUAL DATA STREAMS

DPDP Act 2023 & HIPAA Sandbox

Static Historical EHR:

- 1,000 Synthea Indian Profiles
- Asian Indian BMI (>=23 kg/m²)
- Baseline HbA1c (5.7 - 14.2%)
- Fasting Blood Glucose (FBG)
- eGFR & Serum Creatinine
- Triglycerides / HDL Profile
- Comorbidities & TCF7L2 allele
- Baseline Pharmacotherapy

Dynamic IoT Telemetry:

- 60,120 Timepoints (5-min)
- Interstitial Glucose (CGM)
- Continuous Heart Rate (PPG)
- HRV (RMSSD Autonomic tone)
- Step Cadence & Sleep Stages
- Indian Meal Carbs & Insulins

2. MULTIMODAL FUSION

Cross-Domain Feature Synthesis

Temporal Dynamic Kinetics:

- Glucose Velocity dG/dt (15/30m)
- Glycemic Acceleration d²G/dt²
- Rolling Mean & Std (1h / 3h)
- Intraday CV% Volatility

Autonomic Coupling:

- Autonomic Stress: HR / HRV
- Pre-hypo Sympathetic Surge
- 1-Hour Step Accumulation
- 2-Hour Carbohydrate Load

Latent Patient Embedding:

- Asian Indian Metabolic Profile
- Trigonometric Circadian Wave
- Beta-Cell Reserve Proxy

★ 42 Multimodal Features

3. DUAL-HEAD AI CORE

Prediction & In-Silico Engine

Head 1: 2h Glucose Forecaster

- HistGBM Multi-Tree Regressor
- 2-Hour Horizon (+120 mins)

★ MARD: 7.15% (Gold Std <10%)

★ RMSE: 14.50 mg/dL

- Dynamic 95% Confidence Band

Head 2: Adverse Classifier

- Severe Postprandial Hyper (>200)

  Precision = 65.31%
- Nocturnal Hypoglycemia (<70)

  Early Warning: 120 mins

In-Silico What-If Simulator

- Post-Meal Walk (0-45 mins)
- Carb Swap (-10 to -60g)
- Basal Titration (-4U to +4U)

4. CLINICIAN COCKPIT

Interactive Doctor Dashboard

Clinical Web Interface:

- Virtual Patient Selector
- Static EHR Profile Cards
- 6h Multi-stream Waveforms
- 2h Forward Projection Band
- Dynamic Triage Risk Light

Physician Interactivity:

- Real-time What-If Sliders
- Dynamic Delta Metric Output
- In-Silico Trajectory Overlay
- RSSDI & ADA Decision Support

Explainable AI (XAI):

- Physiological Root-Cause XAI
- Autonomic Crash Alerts

★ FastAPI + Chart.js Web App

DPDP ACT 2023 & HIPAA COMPLIANT: 100% Synthetic & De-identified Data • Zero Real Patient PII • In-Silico Testing Validated